

# SeedHammer II — operator journey

Backing up a 5-of-12 multisig from files, end to end: constellation CLIs on the host, the firmware GUI in the emulator, and the plates that come out.

**Test material only — never put funds behind these keys.**

Every seed in this document was derived deterministically from the public string "shibboleth journey cosigner <i>" so the run reproduces exactly. They are public by construction.

## What this run actually was

The *pathological* wallet from the constellation corpus: `wsh(multi(5, @0...@11))` — twelve cosigners, five signatures.

**One correction, measured rather than assumed.**

CORPUS.md calls its own C6 "pathological deeply-nested miniscript" entry a *placeholder*, noting the 8-nested-or\_d form "actually fits single string (45 B)". The 12-key `multi(5,...)` policy used here comes instead from `phase-6-bucket-C.md`, which calls it "reliably >56 bytes bytecode" — no longer true of today's encoder. `md` bytecode reports **13 bytes**, and it encodes to a **single** md1 string: a keyless BIP-388 template does not pay per key. The chunking in this run comes from the **mk1 key cards**, which do carry xpubs — every one of the twelve splits in two.

## Toolchain, as reported by the binaries

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/release/md --version
md 0.13.0
[exit 0]

$ /scratch/code/shibboleth/mnemonic-key/target/release/mk --version
mk 0.13.0
[exit 0]

$ /scratch/code/shibboleth/mnemonic-secret/target/release/ms --version
ms 0.16.0
[exit 0]

$ /scratch/code/shibboleth/mnemonic-engrave/target/release/me --version
me 0.7.0
[exit 0]

$ /scratch/code/shibboleth/mnemonic-engrave/target/release/me-preview --version
me-preview 0.7.0
[exit 0]
```

## The shape of the job

|                        |                                                                   |
|------------------------|-------------------------------------------------------------------|
| Policy                 | <code>wsh(multi(5,@0/&lt;0;1&gt;/*,...,@11/&lt;0;1&gt;/*))</code> |
| Cosigners              | 12, each at <code>m/48'/0'/0'/2'</code> , mainnet                 |
| md1 descriptor strings | 1 (13-byte bytecode)                                              |
| mk1 key-card strings   | 24 (12 keys × 2 chunks)                                           |
| Public plates          | 25                                                                |
| Secret plates          | 1 (ms1 — typed on the machine, never over the wire)               |
| Total plates           | 26                                                                |

Twenty-six plates is the honest cost of a 5-of-12 held this way, and it is the number the operator most needs before starting.

## The input files

---

Everything the operator supplies lives on disk before a single command runs. Contents below are complete and unedited.

### THE WALLET POLICY

**inputs/wallet-policy.txt**

```
wsh(multi(5,@0/<0;1>/*,@1/<0;1>/*,@2/<0;1>/*,@3/<0;1>/*,@4/<0;1>/*,@5/<0;1>/*,@6/<0;1>/*,@7/<0;1>/*,@8/<0;1>/*,@9/<0;1>/*,@10/<0;1>/*,@11/<0;1>/*))
```

### THE TWELVE SEEDS (SECRET — TEST MATERIAL)

**inputs/seeds/cosigner-00.seed**

ribbon relax cave lock library melody release ranch bird st  
one fossil plate

**inputs/seeds/cosigner-01.seed**

more cotton early draw board party trap announce soft spend  
farm output

**inputs/seeds/cosigner-02.seed**

output island tunnel permit ozone myth purse siren city mam  
mal hobby owner

**inputs/seeds/cosigner-03.seed**

ketchup hobby brand attend cherry face shrimp route define  
arch coconut annual

**inputs/seeds/cosigner-04.seed**

supreme twin piano juice oxygen audit various lounge paddle  
flee speed divide

**inputs/seeds/cosigner-05.seed**

spoon eye blush hip grape supreme road blame chaos dwarf or  
ange wool

**inputs/seeds/cosigner-06.seed**

cart inside barely depend domain lecture castle caution lat  
in resource split coast

**inputs/seeds/cosigner-07.seed**

device accident orchard match coast pass upper decade snow  
tumble cube radio

**inputs/seeds/cosigner-08.seed**

apart sustain chronic afraid thought science speed sting st  
ay slim staff giggle

**inputs/seeds/cosigner-09.seed**

token quote smile section lazy slight problem people invest  
network trophy swear

**inputs/seeds/cosigner-10.seed**

wrap category powder stadium flame oval hedgehog type blast  
cricket chunk this

**inputs/seeds/cosigner-11.seed**

place cute page flash submit hard window scheme loyal cattl  
e possible jealous

## The input files, continued

### THE TWELVE COSIGNER KEYS

#### inputs/keys/cosigner-00.xpub

```
# cosigner 0 – origin [ae6647ee/48h/0h/0h/2h]
xpub6F794Lv12fi3eLBDjDYxbVvK5GrvFQrqB8QpYRLbr8dNtmdtoGc2o5K
iM3pLGd41phXxPw4ytcmbwDYmUfeRGuXE7CniQScVXrYnxz7AJE8
```

#### inputs/keys/cosigner-01.xpub

```
# cosigner 1 – origin [ff4bdd8b/48h/0h/0h/2h]
xpub6DyunbVQJAM2fHpJThHa24tAvbJTRZmbouQuUcZMdkrUvBHfo1hHANP
anDXPZfKSMcCxQMCqqgnfFfjPDLPaqahwsgCMzeFsSyy5QFMU4G
```

#### inputs/keys/cosigner-02.xpub

```
# cosigner 2 – origin [b180e226/48h/0h/0h/2h]
xpub6EQfYuy9Hq9NwLNqhkZu4eAwoUdyCvJewRkVW2v2tZKaSgYZziFZdCr
6TqZJdpjeWiHA2REeJNcQNuHrj1bUot7hv2L3pnNCchmAKJi2Aar
```

#### inputs/keys/cosigner-03.xpub

```
# cosigner 3 – origin [7ab0a774/48h/0h/0h/2h]
xpub6EozX1gsda27873mLzSuK5n9nrUxJ4Z9LuNPRDkbQk8KG4riwQEjLjL
ybaddVedqavScZPhdbwgFA5MDRWkMUE7GYCJ2VSurq98Fu1NUm
```

#### inputs/keys/cosigner-04.xpub

```
# cosigner 4 – origin [bf3e6e44/48h/0h/0h/2h]
xpub6EnMa5NHmcfmAodnmaymYzxhVMp7Zg4sCdEbu9tDtCxcgYE1RbXwXt1p
5S2iDtWbsBw6M6AFRgoQxxTwcSVSwkXHMYfYcMiweUFPebdrYpz
```

#### inputs/keys/cosigner-05.xpub

```
# cosigner 5 – origin [c1826fbb/48h/0h/0h/2h]
xpub6EyaTpwePBqW9HfmgEhKPJLATVxGwvLaPgKgY1s3EzBu7o9noT1Shm
i1wC4Y26xJV4N7adyjg4kyJybUUtYmR2e6cp8JBf7QPzDhN9YiS
```

#### inputs/keys/cosigner-06.xpub

```
# cosigner 6 – origin [ecbd6d3c/48h/0h/0h/2h]
xpub6E7qf5omNGhrv4zEp46p7ZMY7NE5TkhINAobBTfLZ7ZLSdmDKAnVSjh
fvj2kpCvqNyRsYLBZMHu3tfgTyzs8s2DGWhKcuocs3NJC1VC8xE3
```

#### inputs/keys/cosigner-07.xpub

```
# cosigner 7 – origin [7aa32b8d/48h/0h/0h/2h]
xpub6ENGkFakvB7BjHcU7CQWRXn7DpHenFuPCe1sadzMC97Zfoa8uVBezdw
V9SfKJjWjwTNXzJuYQI5cjuDzr2QY5GGMJ4x6cez45r6BWC4RFjv
```

#### inputs/keys/cosigner-08.xpub

```
# cosigner 8 – origin [68c883f9/48h/0h/0h/2h]
xpub6Do4EgnUQc9DGwXxircFtpE5NxAqGWYeJUSqbsbftTDFsDvum2A8pj
A53F2jDosDs75jKiatutz9BVTgcvAto5EshMZdxPYUkip6M2YUos
```

#### inputs/keys/cosigner-09.xpub

```
# cosigner 9 – origin [5a3ae6a9/48h/0h/0h/2h]
xpub6FMrrRRrVL1FGGSiE4j9ve2SBSu1gf7CZTrkfeCn1mcQL96dN2URpWL
tQEsOZDMTU9Xk53rLEatEr4N7s367Q8mjXfC6s5X9guj4Pt8vZpi
```

#### inputs/keys/cosigner-10.xpub

```
# cosigner 10 – origin [ee9b5972/48h/0h/0h/2h]
xpub6DdB6ELpa6SXi6Jy4SEnJXhG6AWGcJoXaULBAYscBbLkgWeVw1wQvTY
BWxqBoaPME4dXpJRuhoBH39UkrhrSWkePhkwRGYXx57CHSAGubn
```

#### inputs/keys/cosigner-11.xpub

```
# cosigner 11 – origin [dd7d1881/48h/0h/0h/2h]
xpub6EPABpiuEwhT2nA6YnHL7WnrmUudC2fWJzKgDaN8Gz5nBhedpLkaFty
vAkxeH9Bkam5k3dVxCXQJekKBr4LGVjirDneAMLjZv5eesz8Ft8k
```

## Host step 1 — the descriptor

```
$ cat /scratch/code/shibboleth/mnemonic-engrave/design/journeys/inputs/wallet-policy.txt
wsh(multi(5,@0/<0;1>/*,@1/<0;1>/*,@2/<0;1>/*,@3/<0;1>/*,@4/<0;1>/*,@5/<0;1>/*,@6/<0;1>/*,@7/<0;1>/*,@8/<0;1>/*,@9/<0;1>/*,@10/<0;1>/*,@11/<0;1>/*))
[exit 0]

$ /scratch/code/shibboleth/descriptor-mnemonic/target/release/md encode --group-size 0 wsh(multi(5,@0/<0;1>/*,@1/<0;1>/*,@2/<0;1>/*,@3/<0;1>/*,@4/<0;1>/*,@5/<0;1>/*,@6/<0;1>/*,@7/<0;1>/*,@8/<0;1>/*,@9/<0;1>/*,@10/<0;1>/*,@11/<0;1>/*))
mdlytpqqxpp3zcpydzk0zdt492x zr7r9qx fc
note: stdout is a keyless descriptor template (no keys)
[exit 0]

$ /scratch/code/shibboleth/descriptor-mnemonic/target/release/md bytecode mdlytpqqxpp3zcpydzk0zdt492x zr7r9qx fc
payload-bits: 100
payload-bytes: 13
hex: 22c200182188b0123456789ab0
note: stdout is a keyless descriptor template (no keys)
[exit 0]

$ /scratch/code/shibboleth/descriptor-mnemonic/target/release/md inspect mdlytpqqxpp3zcpydzk0zdt492x zr7r9qx fc
template: wsh(multi(5,@0/<0;1>/*,@1/<0;1>/*,@2/<0;1>/*,@3/<0;1>/*,@4/<0;1>/*,@5/<0;1>/*,@6/<0;1>/*,@7/<0;1>/*,@8/<0;1>/*,@9/<0;1>/*,@10/<0;1>/*,@11/<0;1>/*))
n: 12
wallet-policy-mode: false
mdl-encoding-id: 1c1c0feba95fef40d5a0e871f2463c52
wallet-descriptor-template-id: 726a666305756435b7c52c5b3fc69c41
origins:
  @0: m
  @1: m
  @2: m
  @3: m
  @4: m
  @5: m
  @6: m
  @7: m
  @8: m
  @9: m
  @10: m
  @11: m
wallet-policy-id: f05e8a1c282f7740bbfd902a759b5577
wallet-policy-id-fingerprint: 0xf05e8a1c
note: stdout is a keyless descriptor template (no keys)
[exit 0]
```

## Host step 2 — a key card

One cosigner shown in full; the other eleven are identical in form.

```
$ cat /scratch/code/shibboleth/mnemonic-engrave/design/journeys/inputs/keys/cosigner-00.xpub
# cosigner 0 - origin [ae6647ee/48h/0h/0h/2h]
xpub6F794Lv12fi3eLBDjDYxbV5G6rvFQrb8QpYRLbr8dNtmdtogc2o5KiM3pLGd41phXxPw4ytcmbwDYmUfeRGuXE7CniQScVXrYnxz7AJE8
[exit 0]

$ wc -l /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/mk-encode-raw.txt
24 /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/mk-encode-raw.txt
[exit 0]

$ /scratch/code/shibboleth/mnemonic-key/target/release/mk inspect mk1qpj6vvppqsqhy6nxvwhxv3lwq5zg3vs76cp0whqh5fkzt3xcf63gqx2
q8yj59pn36zmxs9tvvkpfpkahrevs9gswsqp6ka9cq9yje2y2aa79 mk1qpj6vvppg9lsn2zddmpda6rg6cm06z68pesf3vluall5ym8g05e8cg79d06tf3jp2nft
lvs4nd97mc
xpub:                xpub6F794Lv12fi3eLBDjDYxbV5G6rvFQrb8QpYRLbr8dNtmdtogc2o5KiM3pLGd41phXxPw4ytcmbwDYmUfeRGuXE7CniQScVXrY
nxz7AJE8
origin_fingerprint:  ae6647ee
xpub_fingerprint:    8791855f
origin_path:         48'/'0'/'0'/'2'
  component[0]:       48h (hardened)
  component[1]:       0h (hardened)
  component[2]:       0h (hardened)
  component[3]:       2h (hardened)
policy_id_stubs:     726a6663
chunks:              2
  chunk[0]:           long (BCH variant)
  chunk[1]:           regular (BCH variant)
note: stdout is watch-only - public keys only, cannot spend
[exit 0]
```

### ALL TWELVE, AS ENCODED

| #  | fingerprint | mk1 chunk 1/2                                    | mk1 chunk 2/2                                     |
|----|-------------|--------------------------------------------------|---------------------------------------------------|
| 0  | ae6647ee    | mk1qpj6vvppqsqhy6nxvwhxv3lwq5zg3vs76cp0whqh5...  | mk1qpj6vvppg9lsn2zddmpda6rg6cm06z68pesf3vluall... |
| 1  | ff4bdd8b    | mk1qp4pd0ppqsqhy6nxv0l5hhvtq5zg3vs785q05fj45...  | mk1qp4pd0ppf85yqctkaapxmnt07mjglr6zd7lkh5e7j...   |
| 2  | b180e226    | mk1qplg66ppqsqhy6nxvwccpc3xq5zg3vs7wufpu05k6...  | mk1qplg66ppm425ka4j8hp9h6nwzs5z47l0z3rhwnr3r...   |
| 3  | 7ab0a774    | mk1qpshw5ppqsqhy6nxvdatpfm5q5zg3vs74h9vgvx6p...  | mk1qpshw5pprtt7xm7r5n8lejuk9a3um0arv3te69v8v...   |
| 4  | bf3e6e44    | mk1qpe5pjppqsqhy6nxvwlumjyq5zg3vs748e0mw2c5...   | mk1qpe5pjppqc64lyvss0trqzzah3jp0fc68g7rqu6d7...   |
| 5  | c1826fbb    | mk1qpvr8eppqsqhy6nxv0qcyamamq5zg3vs7c3zlcxs6p... | mk1qpvr8eppprggm8xltzc52l03jwc3n4c7f54ehyghvx...  |
| 6  | ecbd6d3c    | mk1qpenc2ppqsqhy6nxv0kt6mfuq5zg3vs7f7drhpl27...  | mk1qpenc2ppe7pvwaksnp3gy5unqvgm32amwm8983003...   |
| 7  | 7aa32b8d    | mk1qpaz27ppqsqhy6nxvda2x2udq5zg3vs7wf5s0sfdg...  | mk1qpaz27ppqdwh7xccvj00tcwywaznfznv05c9lk5hl...   |
| 8  | 68c883f9    | mk1qp63s7ppqsqhy6nxvd5v3qlq5zg3vs7yw09j7cq...    | mk1qp63s7ppvm7vuqdr6rxc6c9c97a7kcaj2jk7tmzgf...   |
| 9  | 5a3ae6a9    | mk1qpm3z9ppqsqhy6nxvddr4e4fq5zg3vs7lzyle54xg...  | mk1qpm3z9ppgcd6hyffmrlt98m2kxa79pdh5uw3y75wy...   |
| 10 | ee9b5972    | mk1qprpsjppqsqhy6nxv0hfkkjtq5zg3vs7p3wwdzd8h...  | mk1qprpsjppem223hve8z8w6epwepftrn50zekhzdum0...   |
| 11 | dd7d1881    | mk1qpn8y4ppqsqhy6nxv0wh6xypq5zg3vs7wwyena47t...  | mk1qpn8y4ppmesaxh7fy50mx7ms70xd0xf6qpd0ghydl...   |

## Host step 3 — the seed, and the line the tool will not cross

```
$ cat /scratch/code/shibboleth/mnemonic-engrave/design/journeys/inputs/seeds/cosigner-00.seed
ribbon relax cave lock library melody release ranch bird stone fossil plate
[exit 0]

$ /scratch/code/shibboleth/mnemonic-secret/target/release/ms encode --phrase ribbon relax cave lock library melody release ranch bird stone fossil plate
ms10e ntrsq zu3dg yngxu pz9fd fxxpd xkfdl fsa39 6kgte j06n7
word count: 12
language: english (BIP-39 checksum valid)
passphrase: not stored in ms1 (record separately if used)
warning: stdout carries private key material (can spend) – redirect or encrypt (e.g. '> file.txt' or '| age -e ...')
[exit 0]
```

### **ms1 is refused on the radio path, by design.**

Feeding the same seed's ms1 string to the NFC converter does not produce a tag:

```
$ /scratch/code/shibboleth/mnemonic-engrave/target/release/me --in /dev/fd/63 --hex
me: refusing to emit ms1 over NFC: ms1 is secret seed entropy and must never be transmitted by radio (RF eavesdropping risk). Enter it by hand on the device: New > Input Seed > CODEX32.
[exit 3]
```

The descriptor, being public, converts without complaint:

```
$ /scratch/code/shibboleth/mnemonic-engrave/target/release/me --in /dev/fd/63 --hex
0329d1012554006d64317974707171787070337a637079647a6b307a6474343932787a7237723971786663fe
[exit 0]
```

## Host step 4 — validate the bundle, get the checklist

```
$ head -3 /scratch/code/shibboleth/mnemonic-engage/design/journeys/out/backup-strings.txt
mdlytpqqxpp3zcpdydk0zdt492x7r9qxfc
mk1qpj6vvppqsgqhy6nxvhw3lw5zg3vs76cp0whqh5fkzt3xc63gqx2q8yj59pn36zmxs9tvvkpfpkahr9sgswsqp6ka9cq9yje2y2aa79
mk1qpj6vvppg9lsn2zmdpa6rg6cm06z68pesf3vluall5ym8g05e8cg79d06tf3jp2nftlvs4nd97mc
[exit 0]

$ wc -l /scratch/code/shibboleth/mnemonic-engage/design/journeys/out/backup-strings.txt
25 /scratch/code/shibboleth/mnemonic-engage/design/journeys/out/backup-strings.txt
[exit 0]

$ /scratch/code/shibboleth/mnemonic-engage/target/release/me bundle --in /scratch/code/shibboleth/mnemonic-engage/design/journeys/out/backup-strings.txt --preview /scratch/code/shibboleth/mnemonic-engage/design/journeys/out/plates --png --manifest /scratch/code/shibboleth/mnemonic-engage/design/journeys/out/manifest.json
me: rendered plate 1 → /scratch/code/shibboleth/mnemonic-engage/design/journeys/out/plates/plate-1.png
... (plates 2–25 rendered) ...
```

### THE PLATE CHECKLIST, VERBATIM

```
me: backup needs 26 plates (25 public + msl on device):
plate 1/26 msl policy → push via NFC & engrave
plate 2/26 mk1 [ee9b5972/48'/0'/0'/2'] chunk 1/2 → push via NFC & engrave
plate 3/26 mk1 [ee9b5972/48'/0'/0'/2'] chunk 2/2 → push via NFC & engrave
plate 4/26 mk1 [c1826fbb/48'/0'/0'/2'] chunk 1/2 → push via NFC & engrave
plate 5/26 mk1 [c1826fbb/48'/0'/0'/2'] chunk 2/2 → push via NFC & engrave
plate 6/26 mk1 [7ab0a774/48'/0'/0'/2'] chunk 1/2 → push via NFC & engrave
plate 7/26 mk1 [7ab0a774/48'/0'/0'/2'] chunk 2/2 → push via NFC & engrave
plate 8/26 mk1 [ae6647ee/48'/0'/0'/2'] chunk 1/2 → push via NFC & engrave
plate 9/26 mk1 [ae6647ee/48'/0'/0'/2'] chunk 2/2 → push via NFC & engrave
plate 10/26 mk1 [dd7d1881/48'/0'/0'/2'] chunk 1/2 → push via NFC & engrave
plate 11/26 mk1 [dd7d1881/48'/0'/0'/2'] chunk 2/2 → push via NFC & engrave
plate 12/26 mk1 [ff4bdd8b/48'/0'/0'/2'] chunk 1/2 → push via NFC & engrave
plate 13/26 mk1 [ff4bdd8b/48'/0'/0'/2'] chunk 2/2 → push via NFC & engrave
plate 14/26 mk1 [ecbd6d3c/48'/0'/0'/2'] chunk 1/2 → push via NFC & engrave
plate 15/26 mk1 [ecbd6d3c/48'/0'/0'/2'] chunk 2/2 → push via NFC & engrave
plate 16/26 mk1 [bf3e6e44/48'/0'/0'/2'] chunk 1/2 → push via NFC & engrave
plate 17/26 mk1 [bf3e6e44/48'/0'/0'/2'] chunk 2/2 → push via NFC & engrave
plate 18/26 mk1 [68c883f9/48'/0'/0'/2'] chunk 1/2 → push via NFC & engrave
plate 19/26 mk1 [68c883f9/48'/0'/0'/2'] chunk 2/2 → push via NFC & engrave
plate 20/26 mk1 [5a3ae6a9/48'/0'/0'/2'] chunk 1/2 → push via NFC & engrave
plate 21/26 mk1 [5a3ae6a9/48'/0'/0'/2'] chunk 2/2 → push via NFC & engrave
plate 22/26 mk1 [7aa32b8d/48'/0'/0'/2'] chunk 1/2 → push via NFC & engrave
plate 23/26 mk1 [7aa32b8d/48'/0'/0'/2'] chunk 2/2 → push via NFC & engrave
plate 24/26 mk1 [b180e226/48'/0'/0'/2'] chunk 1/2 → push via NFC & engrave
plate 25/26 mk1 [b180e226/48'/0'/0'/2'] chunk 2/2 → push via NFC & engrave
plate 26/26 msl secret → TYPE ON DEVICE (New > Input Seed > CODEX32); never via this tool
```

## Host step 5 — a systemwide payload

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The public cards can also be handed to the machine as one flash payload at 0x10D00000. The digest is what the operator compares on screen.

```
$ /scratch/code/shibboleth/mnemonic-engrave/target/release/me sysw pack --no-passphrase mdlytpqqxp3zcpdzk0zdt492x7r9qxfc
--out /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/sysw-public.bin
strength: no passphrase — BELOW the threshold
digest:   05f3 a2c0 7528 fbf5 03be b78c 2edf 524c
[exit 0]

$ /scratch/code/shibboleth/mnemonic-engrave/target/release/me sysw show /scratch/code/shibboleth/mnemonic-engrave/design/jou
rneys/out/sysw-public.bin
sealed:   false
pub_len:  36
ct_len:   0
identity: e1af502140b8da818dfb69c5fe380bcb239505fbd0fef621dc967e3eb677fb27
digest:   05f3 a2c0 7528 fbf5 03be b78c 2edf 524c
public record 0: mdl/mkl — confirmed
[exit 0]

$ /scratch/code/shibboleth/mnemonic-engrave/target/release/me sysw wipe --fill zeros --out /dev/null
[exit 0]
```



## Host step 6 — the addresses, and where their origins came from

Everything above proves the bytes survive the card format. This is the other half: that they mean the wallet we intended. **Compare these against your coordinator before engraving anything.**

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/release/md address --template wsh(multi(5,@0/48'/0'/0'/2'</0>1>/*,@1/48'/0'/0'/2'</0>1>/*,@2/48'/0'/0'/2'</0>1>/*,@3/48'/0'/0'/2'</0>1>/*,@4/48'/0'/0'/2'</0>1>/*,@5/48'/0'/0'/2'</0>1>/*,@6/48'/0'/0'/2'</0>1>/*,@7/48'/0'/0'/2'</0>1>/*,@8/48'/0'/0'/2'</0>1>/*,@9/48'/0'/0'/2'</0>1>/*,@10/48'/0'/0'/2'</0>1>/*,@11/48'/0'/0'/2'</0>1>/*)) --key @0=xpub6F794Lv12fi3eLBDjDYxbV5GrrvFQrbqBpYRLbr8dNtmdtgc2o5KiM3pLGd41phXxPw4ytcmbwDYmUfeRGUxE7CniQScVxRyNxx7AJE8 --fingerprint @0=ae6647ee --key @1=xpub6DyubnVQJAM2fhPjThHa24tAvbJTRZmbouQuUcZMdkrUvBHF01hHANPanDXP2fKSMcCQxMCqqgnWFjfpPDLPaqahwsGCMZefS5yg5QFMU4G --fingerprint @1=ff4bddd8b --key @2=xpub6EQfYuy9Hq9NwLNgqhKzu4eAwoUdycVjwRkVW2v2tZKaSgYZziFZdCr6TqZdpjwEiHA2REeJNcQNuHrj1bUot7hv2L3pnNCchmAKJi2Aar --fingerprint @2=b180e226 --key @3=xpub6EozX1gsda27873mLzSuK5n9nrUxJ4Z9LuNPRDKbQk8KG4riwQEljLybarddVedqavScZPhdbwgFA5MDRWkMUE7GYCJ2V5Surq98gFu1NUm --fingerprint @3=7ab0a774 --key @4=xpub6EnMa5NHmcfmAodnmaymYzxbVmp7Zg4sCdEbu9tDtcxgYE1RbXwXt1p5S2iDtWbsBw6M6AFRgoQxxtwcSVSwkXHHMYfYcMiweUFFebdrYpz --fingerprint @4=bf3e6e44 --key @5=xpub6EyaTpwePBqW9HFmqEHKpJLATVxGwvLaPgKgjY1s3EzBu7o9noT1ShmilwC4Y26xJV4N7adyjg4kyJybUUtYmR2e6cp8JBf70PzzDhN9YiS --fingerprint @5=c1826fbb --key @6=xpub6E7qf5omNGhrv4zEp46p7ZMY7NE5TKhiNAobBTfLZ7ZLSDmDKAnVSjhfvj2kpCVqNyRsYLBZMHu3tftGTyzs8s2DGWhKuoc3NJC1VC8xE3 --fingerprint @6=ecbd6d3c --key @7=xpub6ENgKFaKvB7BjHcU7CQWRXn7DpHenFuPcE1sadzMC97Zfoa8uVBezdWV95fKJjWJwTNXzJuyQ15cjuDzr2QY5GGMJ4x6cez45r6BWC4RFjv --fingerprint @7=7aa32b8d --key @8=xpub6Do4EgnUQc9DGWxircFtpE5NxAqGWyeJUSQbxsbfTDfSDvum2A8pjA53F2jDosDs75jKiatutz9BVTgcvAto5EshMZdxPYUkip6M2YUos --fingerprint @8=68c883f9 --key @9=xpub6FMrrRRrVLI1FGGSiE4j9ve2SBSUlgf7CZTrkfeCnlmcQL96dN2URpwLtQEsoZDMTU9Xk53rLEatEr4N7s367Q8mjXfC6s5X9guj4Pt8vZpi --fingerprint @9=5a3ae6a9 --key @10=xpub6DdB6ELpaS5X16jY4SEnjXhG6AWGcJoXaULBAyScBbLkgwEwVw1wQvTYBWxqBoaPME4dXpJruhoBH39UkrhrSWkePhkwRGYXx57CHSAgubn --fingerprint @10=ee9b5972 --key @11=xpub6EPABpiuEwhT2nA6YnHL7WrmUudC3fwJzKgDaN8Gz5nBhedpLkaFtyVakxeH9Bkam5k3dVxXQJekKBr4LGVjirDneAMLjZv5eesz8Ft8k --fingerprint @11=dd7d1881 --chain 0 --count 3
bclqvzqg06a7wah6dny5yqmp7a7h5crjudsmc2tggr4zrp0k3edaen2s8jxms4
bclqvceqvwp7erkgf57t32yp9f2r3kss62vnwhkeg8x59qg4cdwzfwq9cfwvz
bclqv8pzc3uqjhs99fcvpx08yj0tp2ju5cqzuw2hehvus0hwk5dsc6n2e09
note: stdout is watch-only — public keys only, cannot spend
[exit 0]
```

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/release/md address --template wsh(multi(5,@0/48'/0'/0'/2'</0>1>/*,@1/48'/0'/0'/2'</0>1>/*,@2/48'/0'/0'/2'</0>1>/*,@3/48'/0'/0'/2'</0>1>/*,@4/48'/0'/0'/2'</0>1>/*,@5/48'/0'/0'/2'</0>1>/*,@6/48'/0'/0'/2'</0>1>/*,@7/48'/0'/0'/2'</0>1>/*,@8/48'/0'/0'/2'</0>1>/*,@9/48'/0'/0'/2'</0>1>/*,@10/48'/0'/0'/2'</0>1>/*,@11/48'/0'/0'/2'</0>1>/*)) --key @0=xpub6F794Lv12fi3eLBDjDYxbV5GrrvFQrbqBpYRLbr8dNtmdtgc2o5KiM3pLGd41phXxPw4ytcmbwDYmUfeRGUxE7CniQScVxRyNxx7AJE8 --fingerprint @0=ae6647ee --key @1=xpub6DyubnVQJAM2fhPjThHa24tAvbJTRZmbouQuUcZMdkrUvBHF01hHANPanDXP2fKSMcCQxMCqqgnWFjfpPDLPaqahwsGCMZefS5yg5QFMU4G --fingerprint @1=ff4bddd8b --key @2=xpub6EQfYuy9Hq9NwLNgqhKzu4eAwoUdycVjwRkVW2v2tZKaSgYZziFZdCr6TqZdpjwEiHA2REeJNcQNuHrj1bUot7hv2L3pnNCchmAKJi2Aar --fingerprint @2=b180e226 --key @3=xpub6EozX1gsda27873mLzSuK5n9nrUxJ4Z9LuNPRDKbQk8KG4riwQEljLybarddVedqavScZPhdbwgFA5MDRWkMUE7GYCJ2V5Surq98gFu1NUm --fingerprint @3=7ab0a774 --key @4=xpub6EnMa5NHmcfmAodnmaymYzxbVmp7Zg4sCdEbu9tDtcxgYE1RbXwXt1p5S2iDtWbsBw6M6AFRgoQxxtwcSVSwkXHHMYfYcMiweUFFebdrYpz --fingerprint @4=bf3e6e44 --key @5=xpub6EyaTpwePBqW9HFmqEHKpJLATVxGwvLaPgKgjY1s3EzBu7o9noT1ShmilwC4Y26xJV4N7adyjg4kyJybUUtYmR2e6cp8JBf70PzzDhN9YiS --fingerprint @5=c1826fbb --key @6=xpub6E7qf5omNGhrv4zEp46p7ZMY7NE5TKhiNAobBTfLZ7ZLSDmDKAnVSjhfvj2kpCVqNyRsYLBZMHu3tftGTyzs8s2DGWhKuoc3NJC1VC8xE3 --fingerprint @6=ecbd6d3c --key @7=xpub6ENgKFaKvB7BjHcU7CQWRXn7DpHenFuPcE1sadzMC97Zfoa8uVBezdWV95fKJjWJwTNXzJuyQ15cjuDzr2QY5GGMJ4x6cez45r6BWC4RFjv --fingerprint @7=7aa32b8d --key @8=xpub6Do4EgnUQc9DGWxircFtpE5NxAqGWyeJUSQbxsbfTDfSDvum2A8pjA53F2jDosDs75jKiatutz9BVTgcvAto5EshMZdxPYUkip6M2YUos --fingerprint @8=68c883f9 --key @9=xpub6FMrrRRrVLI1FGGSiE4j9ve2SBSUlgf7CZTrkfeCnlmcQL96dN2URpwLtQEsoZDMTU9Xk53rLEatEr4N7s367Q8mjXfC6s5X9guj4Pt8vZpi --fingerprint @9=5a3ae6a9 --key @10=xpub6DdB6ELpaS5X16jY4SEnjXhG6AWGcJoXaULBAyScBbLkgwEwVw1wQvTYBWxqBoaPME4dXpJruhoBH39UkrhrSWkePhkwRGYXx57CHSAgubn --fingerprint @10=ee9b5972 --key @11=xpub6EPABpiuEwhT2nA6YnHL7WrmUudC3fwJzKgDaN8Gz5nBhedpLkaFtyVakxeH9Bkam5k3dVxXQJekKBr4LGVjirDneAMLjZv5eesz8Ft8k --fingerprint @11=dd7d1881 --chain 1 --count 3
bclqzjrecytm6jqdqr0zkgazruh69splrxwep5pxvut3xvvn7ged56q3y5udc
bclqthszlwy77zsvajhds5hdccc7l30qx82ghe3mqe4ekj39qfmm0gsmrwdjw
bclqrxshxvtvkewf7vgjtv9u5zstd6m648c44vmgjzrt5z4mffejateq6u3zm5
note: stdout is watch-only — public keys only, cannot spend
[exit 0]
```

### The origins are not on the card.

This policy's wsh( . . . ) wrapper is *canonical*, so md never demanded an explicit origin and none was given — and a keyless template records none of its own. md inspect on the engraved card reports m for all twelve slots, which is what the block below shows. So the twelve derivation paths used above were read from the cosigner key *files*, not recovered from the card: **an operator holding only these plates would have twelve xpub slots and no idea where any of the keys live.**

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/release/md inspect mdlytpqqxp3zcpydzk0zdt492x7r9qxfcc
template: wsh(multi(5,@0</0>1>/*,@1</0>1>/*,@2</0>1>/*,@3</0>1>/*,@4</0>1>/*,@5</0>1>/*,@6</0>1>/*,@7</0>1>/*,@8</0>1>/*,@9</0>1>/*,@10</0>1>/*,@11</0>1>/*))
n: 12
wallet-policy-mode: false
md1-encoding-id: 1c1c0feba95fef40d5a0e871f2463c52
wallet-descriptor-template-id: 726a666305756435b7c52c5b3fc69c41
origins:
  @0: m
  @1: m
  @2: m
  @3: m
  @4: m
  @5: m
  @6: m
```

```
@7: m
@8: m
@9: m
@10: m
@11: m
wallet-policy-id: f05e8a1c282f7740bbfd902a759b5577
wallet-policy-id-fingerprint: 0xf05e8a1c
note: stdout is a keyless descriptor template (no keys)
[exit 0]
```

## The 25 public plates

Rendered by the me-preview sidebar, which reuses upstream SeedHammer curve math — this is the layout the machine will cut.

md1ytpqqxpp3zcpydzk0zdt492  
xzr7r9qxfc



plate 1 — md1 policy

mk1qprpsjppqsqhy6nxv0hfkk  
jq5zg3vs7p3wwdzd8hzkpwz902  
d9kc5t2eft5hteap  
p0tj8yacjs57essr  
vy0hxp6t5pn2z8rs  
zff50c9m78x



plate 2 — key 0 [ae6647ee] chunk 1/2

mk1qprpsjppem223hve8z8w6ep  
wepftrn50zekhzdum0xadvukdl  
2w5xcwergvxzgsnzw4  
xwhe8mcvt4



plate 3 — key 0 [ae6647ee] chunk 2/2

mk1qpv8eppqsqhy6nxv0qcyma  
mq5zg3vs7c3zlcxs6p0uns78cx  
de6v2uw3duvrdqxn  
egu75utyk4vmypz  
pq6n9vkwypcwaakx  
sndgtq25dl2



plate 4 — key 1 [ff4bdd8b] chunk 1/2

mk1qpv8eppprggm8xltzc52l03  
jwc3n4c7f54ehyghvx8f66egfn  
kunvtj7d4sly9qj250  
ldp8h8lcdq



plate 5 — key 1 [ff4bdd8b] chunk 2/2

mk1qpshw5pqsqhy6nxvdatpfm  
5q5zg3vs74h9qvgx6p5tcelvu5  
ys79h3l6k6p077rf  
p0w7gc6wpgs276z0  
28zcp7thvplyhxt  
yfzjd8dg64c



plate 6 — key 2 [b180e226] chunk 1/2

mk1qpshw5pprtt7xm7r5n8leju  
k9a3um0arv3te69v8vvzha94k  
h8yunz4k2za6scp5ag  
x835dfl1c6



plate 7 — key 2 [b180e226] chunk 2/2

mk1qpj6vvpqsqhy6nxvwhxv3l  
wq5zg3vs76cp0whqh5fkt3xcf  
63gqx2q8yj59pn36  
zmxs9tvvkpfpkahr  
evs9gswsqp6ka9cq  
9yje2y2aa79



plate 8 — key 3 [7ab0a774] chunk 1/2

mk1qpj6vvpvg9l5n2zdmpda6rg  
6cm06z68pesf3vluai5lym8g05  
e8cg79d06tf3jp2nft  
lvs4nd97mc



plate 9 — key 3 [7ab0a774] chunk 2/2

mk1qpn8y4pqsqhy6nxv0wh6xy  
pq5zg3vs7wwyena47twxpez7ws  
m3c4x6qus3sjhqrq  
sv6uczrk2tjugkf8  
c8xe5p5pgpgvfj9r  
yja406lqkfk



plate 10 — key 4 [bf3e6e44] chunk 1/2

mk1qpn8y4ppmesaxh7fy50mx7m  
s70xd0xf6qpd0ghydlfs4knp2k  
4pvspusczzvuj59r425  
n6p7wf73at



plate 11 — key 4 [bf3e6e44] chunk 2/2

mk1qp4pd0pqsqhy6nxv0l5hhv  
tq5zg3vs785q05fj45mfq8sumu  
57v5jergsejg9t3f  
6he476ws0gxzx7x4  
gzlzcnp7upqj0mry  
myl9z29zy0h



plate 12 — key 5 [c1826fbb] chunk 1/2

## The 25 public plates, continued



plate 13 — key 5 [c1826fbb] chunk 2/2



plate 14 — key 6 [ecbd6d3c] chunk 1/2



plate 15 — key 6 [ecbd6d3c] chunk 2/2



plate 16 — key 7 [7aa32b8d] chunk 1/2



plate 17 — key 7 [7aa32b8d] chunk 2/2



plate 18 — key 8 [68c883f9] chunk 1/2



plate 19 — key 8 [68c883f9] chunk 2/2



plate 20 — key 9 [5a3ae6a9] chunk 1/2



plate 21 — key 9 [5a3ae6a9] chunk 2/2



plate 22 — key 10 [ee9b5972] chunk 1/2



plate 23 — key 10 [ee9b5972] chunk 2/2



plate 24 — key 11 [dd7d1881] chunk 1/2



plate 25 — key 11 [dd7d1881] chunk 2/2

**Plate 26 is not here, and cannot be.**

The seed plate is engraved from an ms1 string typed on the machine. No host tool in this constellation will render, transmit or preview it.

## On the machine — finding the program

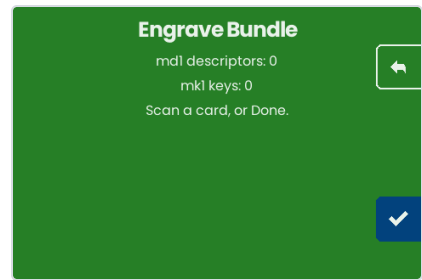
The emulator runs the real firmware GUI compiled to wasm, driven only through the touch input the machine itself has.



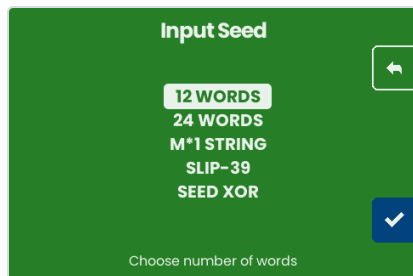
Boot: the program carousel, 10 entries



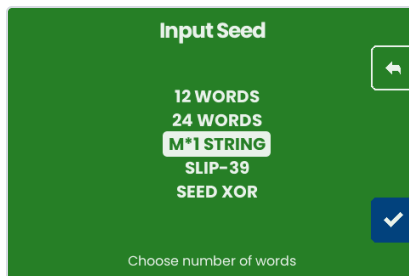
Engrave Bundle — takes md1 + mk1 cards



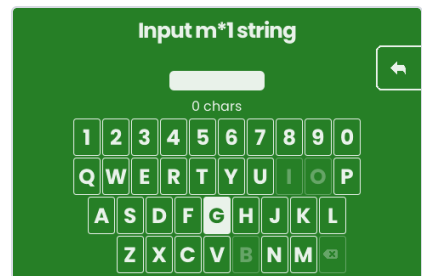
The gatherer, waiting for cards



Backup Wallet: seed source first



M\*1 STRING selected

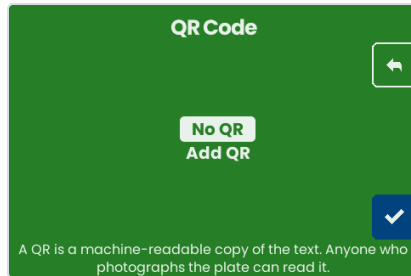


bech32 keyboard — I, O and B are absent

### The 10 programs, read off the carousel:

Backup Wallet · BIP-39 Password · Engrave Text · Account Xpub · Engrave Bundle · Engrave Single-Sig · Engrave Multisig · Load Payload · BIP-85 Child Seed · Sealed Payload.

## On the machine — composing a plate

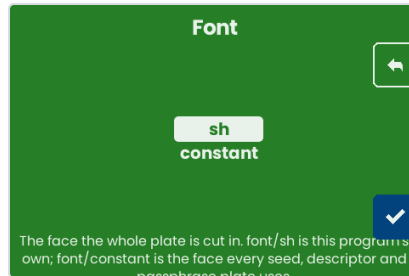


QR Code

No QR  
Add QR

A QR is a machine-readable copy of the text. Anyone who photographs the plate can read it.

QR: the warning is that a photo of the plate is a copy of the text

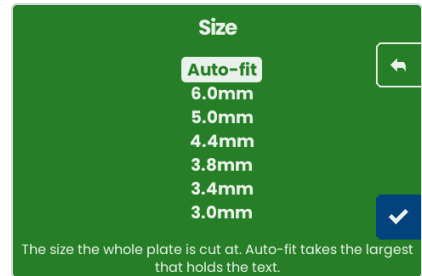


Font

sh  
constant

The face the whole plate is cut in. font/sh is this program's own; font/constant is the face every seed, descriptor and passphrase plate uses.

Font: font/sh is this program's own; font/constant is the face every seed and descriptor plate uses



Size

Auto-fit  
6.0mm  
5.0mm  
4.4mm  
3.8mm  
3.4mm  
3.0mm

The size the whole plate is cut at. Auto-fit takes the largest that holds the text.

Size: auto-fit takes the largest that holds



Text

6.0mm 1/24 lines

q w e r t y u i o p  
a s d f g h j k l  
z x c v b n m  
ABC space show nl

Text entry masks by default



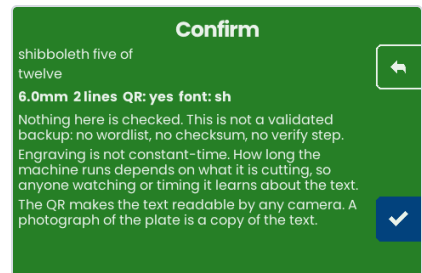
Text

6.0mm 1/24 lines

shibboleth five of twelve

q w e r t y u i o p  
a s d f g h j k l  
z x c v b n m  
ABC space hide nl

"show" reveals it



Confirm

shibboleth five of twelve

6.0mm 2 lines QR: yes font: sh

Nothing here is checked. This is not a validated backup: no wordlist, no checksum, no verify step. Engraving is not constant-time. How long the machine runs depends on what it is cutting, so anyone watching or timing it learns about the text. The QR makes the text readable by any camera. A photograph of the plate is a copy of the text.

Confirm — and it says plainly that nothing here is checked

## On the machine — the engrave



## The plate overlay, live

At the moment the engrave begins the emulator draws the **whole planned layout in grey**, from the same PlanEngraving call the firmware makes. What the driver has actually stepped is filled in **black**, decoded from the real step stream — not from the plan. The red circle is the head.



### Why the last frame reads 0.0, 0.0.

The job ends by homing, so plan and progress stay in one coordinate space. Before F-121 was fixed the emulator did not home while the device did, and a resumed cut drew its progress roughly 68 mm away from its own plan.

## What this run turned up

---

### 1. The emulator freezes when an NFC tag is presented to a gathering flow.

`bundleGatherFlow` samples `Platform.NFCReader()` once on entry, so a tag must already be pending or no scanner goroutine starts at all — that part is merely awkward. The defect is what happens when one *is* pending: the emulator's source is one-shot, so after the single record is read the reader sits at EOF, and the scan loop only sleeps on `scanFailed`. On `io.EOF` it spins. Under Go/wasm, which is cooperatively scheduled on the browser's single thread, a goroutine that never blocks starves the event loop and the page stops responding — confirmed here: after presenting a card, even navigating away from the tab timed out.

The consequence is that the NFC path added for the systemwide-payloads work — the one path spec §8.2 exists to make walkable — cannot currently be walked in the emulator, and a 25-card bundle could not be delivered over it in any case, since one flow entry can consume exactly one tag.

### 2. CORPUS.md's C6 is still a placeholder, and the 12-key policy is no longer chunk-forcing.

Measured: 13 bytes of bytecode, one md1 string. Whatever the corpus wants C6 to demonstrate, this policy no longer demonstrates it — though it remains an excellent *operator* stress case at 26 plates.

### 3. `me bundle --preview` requires an empty output directory

and exits 2 otherwise. Correct behaviour — worth knowing before a re-run.

## Reproducing this document

---

```
cd <this directory>
bash transcript.sh > transcript.txt 2>&1      # every CLI block
python3 build_pdf.py                          # this PDF

# the cosigner set (run inside the seedhammer flake)
go run ./cmd/journeykeys > keys.json
```

---

The emulator frames were captured by `POSTing` `canvas.toDataURL()` to a local receiver, so each screenshot is the device framebuffer exactly — no browser chrome, no scaling.